Persistent plans under an exact simulator

Confirmation run · clean working tree
2026-09-05 10:01 UTC · 240 recorded runs

Abstract

We compared persistent action-plan populations with fresh evolution, random shooting and categorical CEM using exact dynamics in a unit-square puck task. Each run contained 80 actions; the protocol declared 30 paired seeds per environment. In open, observed successes were Persistent RTGA 30/30; Fresh evolution 30/30; Random shooting 30/30; Categorical CEM 30/30. In rooms, observed successes were Persistent RTGA 26/30; Fresh evolution 28/30; Random shooting 26/30; Categorical CEM 27/30. These tests isolate search behavior with a supplied objective. They do not test learned dynamics, curiosity or transfer.

Question: does keeping a living population improve navigation when the simulator and objective are already known?

This is the planner diagnostic. Learned models and autonomous curiosity are separate experiments.

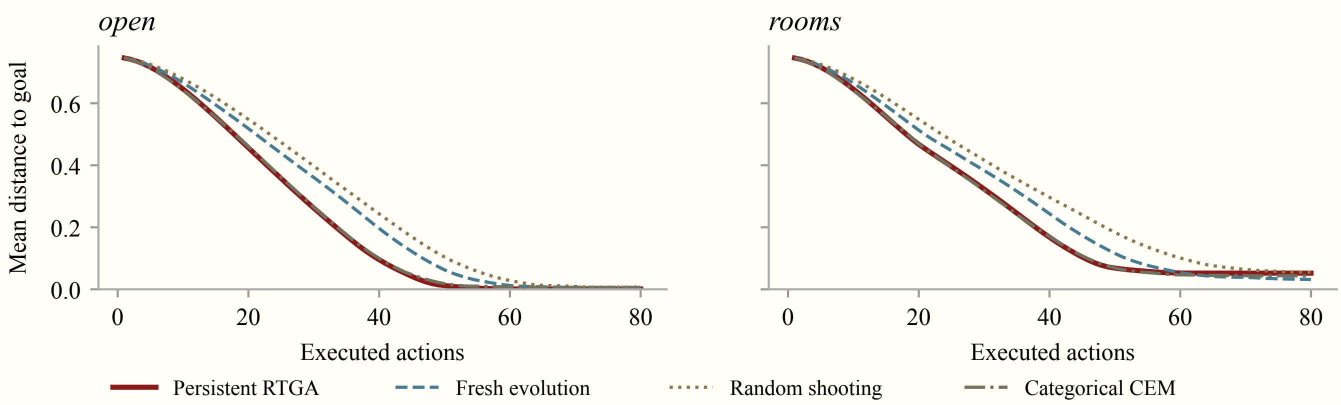

Figure 1. Distance after each executed action, averaged equally across recorded seeds. Lines are descriptive means; paired intervals are on the next page.

Recorded outcomes and planning cost

Variant / method	*Success*	*First hit†*	*Mean dist.*	*Final dist.*	*Latency p50 / p95, ms*
open / Persistent RTGA	30/30	42.5	0.226	0.001	3.50 / 3.91
open / Fresh evolution	30/30	49.5	0.270	0.006	3.51 / 3.97
open / Random shooting	30/30	53.5	0.293	0.008	3.24 / 3.66
open / Categorical CEM	30/30	43.5	0.230	0.005	3.38 / 3.80
rooms / Persistent RTGA	26/30	45.0	0.268	0.052	8.39 / 8.91
rooms / Fresh evolution	28/30	51.5	0.296	0.031	8.39 / 8.90
rooms / Random shooting	26/30	58.5	0.332	0.055	8.13 / 8.65
rooms / Categorical CEM	27/30	45.0	0.265	0.044	8.26 / 8.77

Persistence did not dominate the observed success counts: Fresh evolution reached more goals than persistence in rooms (28/30 versus 26/30); Categorical CEM reached more goals than persistence in rooms (27/30 versus 26/30). Improved distance control and reliable goal reachability are separate endpoints.

† First hit is the median among successful runs only. Mean/final distance include all runs. p50/p95 pool decision latencies within the row.

Recorded per-decision model-transition counts: 6,144. The suite contains 117,964,800 hypothetical transitions. Latency percentiles pool recorded decisions within each method/variant, rather than averaging run percentiles. Statistical intervals use seeds, not decisions, as independent units.

All methods execute one action per decision. Re-evaluating inherited plans is charged to the same model-transition budget.

Paired differences: persistent minus comparator

Variant / comparator	*Δ success, pp*	*Δ restricted hit*	*Δ mean distance*	*Δ final distance*
open / Fresh evolution	+0.0 [+0.0, +0.0]	-7.4 [-8.5, -6.3]	-0.044 [-0.051, -0.038]	-0.005 [-0.007, -0.004]
open / Random shooting	+0.0 [+0.0, +0.0]	-11.1 [-12.3, -9.9]	-0.067 [-0.076, -0.059]	-0.007 [-0.008, -0.005]
open / Categorical CEM	+0.0 [+0.0, +0.0]	-0.5 [-0.8, -0.1]	-0.003 [-0.005, -0.002]	-0.004 [-0.005, -0.003]
rooms / Fresh evolution	-6.7 [-16.7, +0.0]	-5.2 [-7.7, -2.0]	-0.028 [-0.042, -0.010]	+0.021 [-0.005, +0.058]
rooms / Random shooting	+0.0 [-10.0, +10.0]	-10.9 [-13.3, -8.5]	-0.064 [-0.079, -0.049]	-0.003 [-0.035, +0.032]
rooms / Categorical CEM	-3.3 [-10.0, +0.0]	+0.3 [-0.7, +1.9]	+0.003 [-0.003, +0.013]	+0.008 [-0.005, +0.035]

Every effect is the mean paired difference, persistent minus comparator, within one variant. The 95% percentile intervals resample complete seed pairs 20,000 times (NumPy generator seed 20260905). Success effects are percentage points. Distance effects use world units. Restricted first-hit time is min(T, 80); failures contribute 80 actions. Negative distance/time differences favor persistence. Intervals are unadjusted across comparisons and endpoints; degenerate success intervals do not prove equal success probabilities.

Read the endpoints separately. More successes favor persistence; smaller distance and first-hit time favor persistence.

Each displayed interval contains 95% of bootstrap estimates. No multiplicity correction.

Methods

The agent selects among six primitive actions, simulates 64 plans of 24 steps for 4 generations, executes the best first action, then replans. Fitness is negative discounted mean future distance (discount 0.99); success means observed distance ≤ 0.06 at least once. Open has no internal wall; rooms has a permanent open doorway. Initial positions and goals are matched by seed.

Persistent RTGA shifts all genomes and appends a random action. Fresh evolution uses the same genetic operators but initializes anew at each decision. Random shooting samples independent populations; categorical CEM refits per-position action probabilities with a 10% uniform floor and keeps the incumbent within each decision. CEM is initialized afresh between decisions, so this is not an iCEM comparison.

Limits of this result

This is one simple environment family, one selected compute setting and known deterministic dynamics. The goal-directed distance objective provides dense guidance, and rooms does not require discovering a switch. Reaching the goal once is not the same as remaining there; mean and final distance expose that difference. The CEM baseline lacks cross-decision warm starts. Planning-time measurements exclude real environment steps, trace writing and training; they are not hard real-time guarantees. No learned-model or curiosity claim follows.

Related work: RHEA shift buffers (2017) and iCEM (2020). Categorical CEM here is an intentionally simple comparator.

Provenance

Run commit `41bd35db6b875376d5f9a3557c4f4029c965042c` · clean working tree. The run began from a clean recorded commit. The result file also contains per-source SHA-256 hashes.
Results ../results/001-oracle/results.json
Result SHA-256 `a44d9f99c4c21dd7f53019823fe551014b2cd9b71cc4f9a45c21dc71ffa4fddc`
Seeds 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129 · Python 3.12.11 · NumPy 2.5.2
macOS-26.6.2-arm64-arm-64bit
Recorded invocation:

```
python -m rtga.cli oracle --output results/001-oracle --seeds 30 --seed-start 100 --steps 80 --population 64 --
horizon 24 --generations 4
```

The Markdown companion records the report-generation command. Numerical results are derived from one complete JSON snapshot; this paper does not select a favorable animation.